

Edge AI Empowered UAV with Onboard Semantic Extraction for Monitoring Applications

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Abstract—UAV platforms can provide new services on the move that go beyond pure communication. This work integrates remote sensing, embedded processing, and 5G connectivity on a UAV platform as an end-to-end intelligent monitoring solution. The system leverages RGB and multispectral (MS) sensors to execute AI tasks (i.e., object detection, MS-RGB fusion, semantic segmentation, and spectral index extraction) directly onboard. Processed telemetry and AI outputs feed a high-fidelity Digital Twin for operational replication and synthetic data generation. Experimental results in the field confirm that the integrated edge AI approach significantly reduces bandwidth requirements by 5x, 8x, and 10x, respectively, for semantic segmentation, MS indexing, and object detection, thus allowing efficient monitoring even under low connectivity and power limitations.

Index Terms—UAV, Edge AI, Digital Twin, Multispectral Imaging, Bandwidth Optimization, Computer Vision, 5G.

I. INTRODUCTION

Unmanned Aerial Vehicles (UAVs) are increasingly adopted for intelligent monitoring applications thanks to their mobility, rapid deployment, and ability to integrate sensing, communication, and onboard processing capabilities. By exploiting 5G terrestrial connectivity or Low Earth Orbit (LEO) satellite links, UAVs can provide Internet access and remote sensing services even in infrastructure-limited environments. These characteristics make UAVs particularly attractive for smart agriculture, environmental monitoring, and emergency management applications [1], [2].

In this context, the Horizon Europe 5G-HUB project investigates integrated terrestrial and non-terrestrial networking solutions for mission-critical UAV-assisted services [3]. However, the implementation of such systems remains challenging due to strict payload and energy constraints, UAV mobility, real-time onboard processing requirements, and the need for efficient TN/NTN integration.

Recent works have explored UAV-based edge intelligence for environmental monitoring and smart agriculture. UAV platforms combined with MEC and IoT infrastructures have been proposed for CO₂ monitoring and decision support [4], although AI processing is typically offloaded to external servers. Other approaches demonstrated onboard object detection and semantic segmentation using embedded GPU

platforms [5]–[7], as well as hyperspectral or multispectral analysis for agricultural monitoring [8], [9]. Additional systems integrated edge AI with 4G/5G connectivity, multimodal sensing, or contextual analysis for emergency scenarios [10]–[12]. However, existing solutions generally focus on single AI tasks or sensing modalities, lack quantified communication-bandwidth reduction, or do not support integrated Digital Twin synchronization and multispectral semantic analysis under realistic UAV constraints.

This paper presents a demonstration activity on edge AI-enabled UAV monitoring developed within the NETWIN project of the Italian RESTART framework [13]. The proposed system implements an end-to-end intelligent UAV monitoring chain integrating RGB and multispectral sensing, onboard AI processing, 5G connectivity, and Digital Twin synchronization. The platform supports real-time YOLO-based object detection, semantic segmentation, multispectral index extraction, and MS-RGB image fusion directly onboard the UAV. By processing high-dimensional data locally before transmission, the proposed edge AI approach significantly reduces communication overhead while enabling efficient monitoring in bandwidth-constrained scenarios.

This paper is organized as follows: Section II describes the system architecture and the algorithmic approach; Section III details the edge AI hardware and software implementation; Section IV provides the experimental results, followed by the concluding remarks in Section V.

II. SYSTEM ARCHITECTURE AND DESIGN

The proposed UAV monitoring system operates under several key assumptions: (i) limited or unreliable TN connectivity in the target operational areas, (ii) onboard processing autonomy, (iii) stringent power and weight constraints, (iv) real-time or near-real-time responses to field conditions, and (v) high-fidelity data logging and post-flight analysis.

A. System Architecture

The communication architecture of the proposed UAV platform exploits multiple technologies to ensure reliable data transmission. In particular, the UAV acts as a mobile gateway,

TABLE I
COMPARISON OF UAV-BASED EDGE MONITORING SYSTEMS.

System	Processing Unit	Sensors	Tasks	Communication	BW Reduction / Digital Twin
Tilon et al. [5]	Embedded GPU (Jetson)	RGB	Detection, Segmentation	4G LTE / Wi-Fi	Not quantified / No
Hernandez et al. [6]	Jetson family (GPU)	RGB	Semantic segmentation	Not specified	$\sim 6\times$ (83.4%) / No
AERO [7]	Jetson Xavier NX	RGB	Detection + Tracking	4G/5G/Wi-Fi	Not quantified / No
ETAUS [12]	FPGA (Zynq UltraScale+)	RGB (HSI conv.)	AQI classif. + De-identification	Wireless	Not quantified / No
Navarro et al. [10]	Jetson Orin Nano	360° RGB	Detection + LLM analysis	4G	Not quantified / No
Bultmann et al. [11]	Intel NUC + EdgeTPU + iGPU	LiDAR + RGB + Thermal	Segmentation + Detection	None spec.	Not quantified / No
Manoj et al. [9]	Jetson Nano	RGB + Multispectral + IoT	Crop disease detection	Wireless	Not quantified / No
This work	Jetson Orin Nano	RGB + Multispectral	Detection, Segm., NDVI/NDRE, Fusion	5G/NTN	5–10$\times$ / Yes

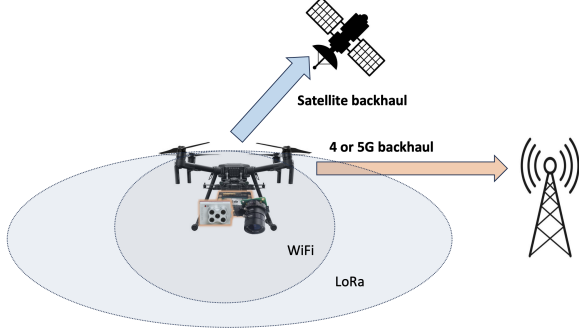


Fig. 1. UAV extending communication area through 5G or satellite backhaul.

leveraging a high-speed 5G backhaul to provide Internet connectivity to onboard devices, while Wi-Fi supports short-range communication with nearby ground terminals (see Figure 1). Moreover, the architecture can be extended to support LoRa-based IoT devices through a lightweight onboard gateway [14]. In areas without terrestrial cellular coverage, satellite connectivity can also be integrated into the platform, as shown in [15] and investigated in the 5G-HUB project [3].

Onboard edge computing plays a key role in enabling resource-efficient communications. By processing data locally before transmission, the system significantly reduces traffic volume and mitigates the throughput limitations of constrained backhaul links. This makes satellite-assisted monitoring feasible even in remote or emergency scenarios.

The sensing pipeline integrates an RGB camera and a multispectral (MS) sensor acquiring five GeoTIFF bands (Blue, Green, Red, Red-Edge, and Near-Infrared). To ensure consistency under varying illumination conditions, radiometric calibration is performed using a calibrated reflectance panel together with the integrated sunshine sensor, following the workflow in [11]. RGB and MS acquisitions are synchronized, while the processed outputs are stored locally and transmitted to a remote server through 5G TN/NTN connectivity.

B. Algorithmic Framework

This subsection describes the onboard image-processing methods adopted by the UAV.

1) *Object Detection*: The system adopts a lightweight YOLOv11-based detector for real-time onboard object detection. The model is trained using data augmentation techniques

based on the VisDrone dataset [16], enabling adaptation to different environmental monitoring scenarios. The lightweight design allows real-time inference directly onboard while maintaining reduced computational complexity.

2) *Semantic Soil Segmentation*: The proposed framework implements semantic soil segmentation to classify different soil typologies. A patch-based approach is adopted, where each input image is decomposed into smaller sub-regions that are independently analyzed.

3) *Semantic Labeling*: Semantic labeling is implemented through a semi-supervised pipeline. Initially, an unlabeled dataset is partitioned using K -means clustering to identify representative classes such as “Healthy Vegetation”, “Bare Soil”, and “Others”. These preliminary labels are refined through field observations to train a lightweight ResNet-18 classifier [17], optimized for patch-level classification. The individual patch predictions are then reconstructed into a spatial heatmap, providing a semantic representation of the monitored area.

4) *NDVI/NDRE Indices*: MS reflectance values are used to compute two vegetation indicators: the Normalized Difference Vegetation Index (NDVI), which evaluates biomass and vegetation vigor [18], and the Normalized Difference Red Edge (NDRE), which improves vegetation analysis in dense canopies [19]. These indices are defined as

$$\text{NDVI} = \frac{\text{NIR} - \text{RED}}{\text{NIR} + \text{RED}} \quad \text{and} \quad \text{NDRE} = \frac{\text{NIR} - \text{RE}}{\text{NIR} + \text{RE}}. \quad (1)$$

Here, RE, NIR, and RED denote the Red-Edge, Near-Infrared, and Red spectral bands, respectively. NDVI values above 0.3 generally indicate healthy vegetation, while NDRE values above 0.4 correspond to high chlorophyll content. The indices are computed using NumPy and Rasterio libraries, enabling the generation of high-resolution vegetation heatmaps for monitoring applications.

5) *Image Fusion Framework*: To overcome the trade-off between spatial and spectral resolutions of the onboard cameras, we adopt a regression-based image fusion framework [20], [21]. While the MS sensor provides spectral information required for vegetation analysis, the RGB camera offers significantly higher spatial resolution. The proposed framework injects high-frequency spatial information from RGB imagery into the multispectral bands [22], generating fused outputs that preserve both fine spatial details and spectral characteristics.

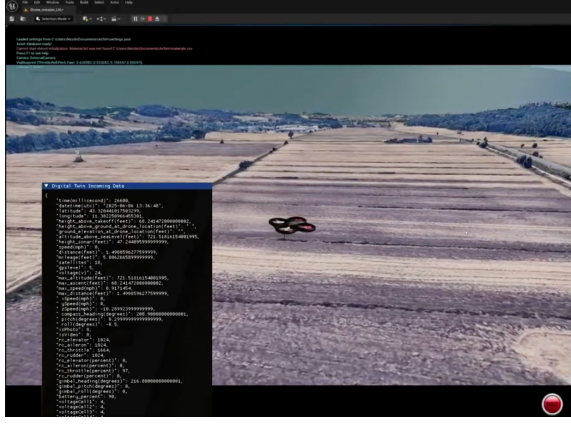


Fig. 2. An example of the replica of a UAV mission.

This improves the interpretability of vegetation regions and supports downstream monitoring tasks.

C. Digital Twin Integration

The Digital Twin (DT) framework enables high-fidelity replication of UAV missions together with synthetic data generation. Our implementation uses DITTO for data management, behavioral models for UAV replication, and Unreal Engine to render both the UAV and the surrounding 3D environment. Terrain information is reconstructed from Google Maps data according to the UAV location.

During the mission, the physical UAV continuously transmits telemetry and status information through a 5G backhaul to the DT platform, enabling real-time visualization and post-flight analysis. The DT framework also supports mission replay using previously recorded telemetry traces. In addition, synthetic datasets can be generated by introducing additional virtual objects into the reconstructed scene (see Figure 2), supporting ML/AI data augmentation and validation activities.

III. AI EDGE PROCESSING: HARDWARE AND SOFTWARE IMPLEMENTATION

This section details the practical implementation of the edge-computing and communication payload.

A. Hardware Configuration

Our UAV has a payload whose electronics was driven by the constraints of a maximum payload weight of 1.6 kg and power limitation to 40-50 W. Our payload configuration (see Figure 3) integrates a high-resolution Raspberry Pi RGB camera (12.3 MP) for contextual imaging and a MicaSense RedEdge-MX MS sensor. Processing is handled by a modular onboard computing unit that supports either a Raspberry Pi 4 Model B for basic tasks or an NVIDIA Jetson Orin Nano GPU with CUDA-accelerated AI. Reliable telemetry and communication with a remote server is ensured via a dual Wi-Fi/5G modem-router for onboard connectivity and data backhaul. It can also provide connectivity to close ground Wi-Fi devices. To maintain optical alignment and mitigate airframe vibrations, a



Fig. 3. UAV payload.

custom 3D-printed mount co-locates the sensors. The system is powered by a dedicated 27,000 mAh / 140 W power module, which provides regulated multi-voltage rails (5–20 V), independent from the flight battery to preserve operational endurance.

B. Software Stack and Telemetry Acquisition

The software architecture is developed in Python v3.8+, employing a specialized suite of libraries for edge computing and data analysis. Fundamental numerical operations and data manipulation are performed via NumPy and Pandas [23], while Scikit-learn and PyTorch [24] serve as the primary engines for machine learning and deep learning model development. This stack is optimized for inference on the adopted resource-constrained edge devices. Real-time acquisition from RGB and MS sensors is managed through OpenCV [25], ensuring efficient frame synchronization, while the Weights & Biases (W&B) platform is integrated for systematic experiment tracking. Moreover, during the flight mission, all relevant telemetry and system metrics are continuously acquired, including UAV flight dynamics (e.g., airspeed, GPS coordinates, relative and absolute altitude, roll, pitch, and yaw angles), power subsystem status (e.g., battery level and voltage), and onboard processing unit performance indicators (e.g., CPU and GPU utilization, memory footprint, and power consumption). These data streams are recorded and then transmitted to the DT platform, enabling high-fidelity replication and analysis of the UAV's operational state.

IV. NUMERICAL RESULTS

This section presents the experimental results of field tests conducted in a predefined rural area near Siena, Italy. First, we performed a UAV flight mission, capturing simultaneous MS and RGB images of the scene, including objects such as vehicles and agricultural fields. The acquired images were then processed onboard the UAV using different algorithms as detailed below.



Fig. 4. Object detection.

A. Image Processing Tasks

This subsection reports the methodology and the results achieved by onboard image-processing for object detection, vegetation-index mapping, semantic soil segmentation, and MS-RGB fusion. Quantitative metrics are provided for semantic segmentation, while the other tasks are evaluated qualitatively through visual inspection of the onboard-generated outputs as illustrated in next Figures 4–8.

Object detection uses a YOLOv11 backbone fine-tuned in VisDrone-2019 ($\sim 10k$ aerial images with 9 object classes, including pedestrians, vehicles, and bicycles). The model uses a 640×640 input and a 102 MB checkpoint, and is deployed on the UAV only for onboard inference, without any training or weight updates during the mission. Field testing on the Siena RGB stream is qualitative, as shown in Figure 4. Although object detection tests addressed only vehicles due to the practical availability of cars in the test environment (see Figure 4), the system has been designed to be dataset-agnostic and can be re-trained on domain-specific datasets for alternative use cases, such as wildlife or human monitoring.

MS-RGB fusion co-registers the five MicaSense bands to the synchronously-triggered 12.3 MP RGB via phase cross-correlation and injects the RGB high-frequency details into each MS band by robust regression [20], [21]. Figure 5 shows an application of the image fusion task to a vegetation area. This fused image enhances vegetation boundaries and preserves fine branch structures, improving the representation of vegetation regions. Quantitatively, the per-band sampling increases from MicaSense 1280×960 (~ 1.23 MP) to RGB 4056×3040 (~ 12.3 MP, $\sim 10\times$ area gain), while preserving all five MS channels. The validation is performed visually and by checking that NDVI/NDRE values computed on the fused product remain within the same ranges as Figure 6.

Regarding **vegetation index mapping (i.e., NDVI and NDRE)**, the output of the acquired area is shown in Figure 6. This type of graphical representation allows us to identify areas where vegetation is healthy ($\text{NDVI} > 0.3$) and with high chlorophyll ($\text{NDRE} > 0.4$).

For **semantic soil segmentation**, the ResNet-18 classifier is trained on 12 500 patches from five reference RGB images of the surveyed area, corresponding to 38.2 MB of raw data. Each image has an approximate resolution of 2950×5250 pixels and is split into a 50×50 grid, producing 2500 non-overlapping patches per image. Each extracted patch is then

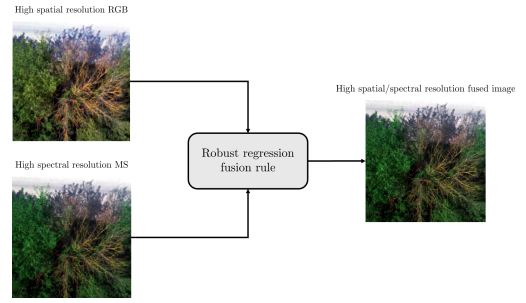


Fig. 5. Schematic illustration of MS-RGB image fusion, applied to a crop from the image shown in Figure 6.

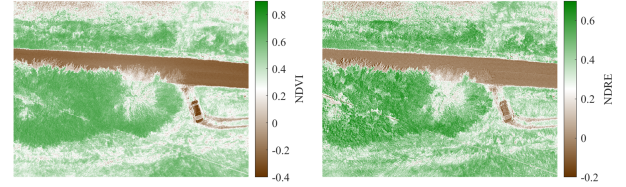


Fig. 6. NDVI/NDRE calculations.

resized to 128×128 pixels. The model is deployed locally on the edge device, occupies 44.79 MB, and no weights are transmitted during the mission. Patch labels are assigned semi-automatically using a GUI-driven K-means tool with $k = 4$, following the four semantic classes in Sect. II-B2. The model is initialized from ImageNet and trained with Adam using learning rate 3×10^{-4} , weight decay 10^{-4} , batch size 32, $\pm 10^\circ$ rotations, random flips, and early stopping based on validation loss. The validation uses 10 000 patches from four independent flights at 30/50/60/120 m altitudes (see Table II), enabling cross-flight evaluation under unseen acquisition conditions. Because the classes are imbalanced, with “Others” accounting for only $\sim 3\%$ of the validation patches, we report per-class precision, recall, F1-score, and support (i.e., the number of validation patches in each class) in Table II, while overall accuracy is reported below as a global metric over all validation patches. Under cross-flight validation, the ResNet-18 reaches 85.9% overall validation accuracy, computed as the ratio of correctly classified patches to the total number of validation patches (corresponding to macro F1 = 0.82, and weighted F1 = 0.86), with training accuracy of 95.6% at the early-stopped epoch. Per-class scores in Table II and Figure 7 show strong performance for “Healthy Vegetation” and “Bare Soil” (F1 = 0.90 and 0.92), while “Stressed Zone” remains the most challenging class (F1 = 0.55), mainly due to confusion with healthy or bare patches at higher altitudes. All scores are computed directly from the model predictions on the validation patches, without manually adjusting the decision rules after training. Figure 8 shows the semantic soil segmentation map obtained from patch-level classification.

B. Runtime, Resource Usage, and Bandwidth Analysis

We next evaluate whether the onboard pipeline fits the runtime, energy, memory, and bandwidth constraints of the

TABLE II
PER-CLASS PERFORMANCE OF THE RESNET-18 PATCH CLASSIFIER ON THE 10 000 VALIDATION PATCHES.

Class	Precision	Recall	F1	Support
Healthy Vegetation	0.93	0.87	0.90	4474
Bare Soil	0.93	0.91	0.92	3975
Stressed Zone	0.50	0.61	0.55	1242
Others	0.83	0.98	0.90	309
Macro avg	0.80	0.84	0.82	10000
Weighted	0.87	0.86	0.86	10000

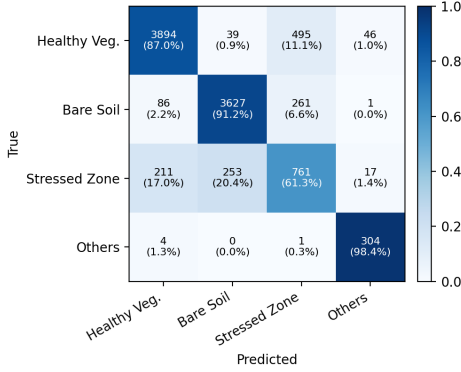


Fig. 7. Confusion matrix on the four validation flights (30, 50, 60, 120 m).

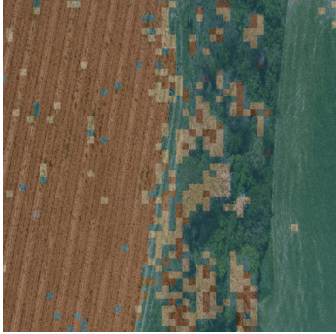


Fig. 8. Semantic segmentation.

UAV payload. In particular, we evaluate the two payload options in Section III-A: the CPU-based Raspberry Pi 4B and the GPU-based Jetson Orin Nano. Table III reports the per-task measurements. The Pi 4B completes one round (i.e., one processing cycle, defined as the sequential execution of all onboard tasks on one synchronized image set composed of one RGB image and five MS band images) in 302.97 s, mainly due to MS–RGB fusion, which accounts for about 88% of the runtime. Object-detection inference takes 2.47 s, with 102/255 ms for pre-/post-processing, while the segmentation pass adds about 2 s for 2500 patches. For a complete *processing round*¹, the Pi 4B consumes 0.344 Wh, uses 2.22 GB peak RAM, and

¹Intending for round, a processing cycle with object detection, image fusion, vegetation indexes computation, and segmentation computed on the same set of images.

TABLE III
PER-TASK ONBOARD INFERENCE COST FOR THE RESNET-18 SETUP IN A SINGLE ROUND. RUNTIME AND PEAK RAM VALUES ARE MEASURED DIRECTLY ON THE CORRESPONDING ONBOARD PLATFORMS.

Task	Pi 4B (s)	Orin Nano (s)	Peak RAM (Pi/Orin) (MB)	Notes
RGB acquisition	1.64	1.64	635.6 / 632.0	I/O bound
Soil segmentation	2.00	0.67	642.0 / 605.0	ResNet-18, 2500 patches
Object detection	8.97	0.28	644.7 / 540.0	TensorRT FP16 on Orin
NDVI/NDRE indices	22.68	1.95	638.5 / 615.0	numpy / rasterio
MS–RGB image fusion	267.68	24.70	648.1 / 720.0	Torch+CUDA on Orin
Total	302.97	29.24	—	

Pi round-level: 0.344 Wh, 2.221 GB RAM, CPU 39.4 °C.
Orin round-level: 1.55 Wh, 1.47 GB RAM, GPU 53.7 °C.

reaches 39.4 °C CPU temperature. The Orin Nano completes the same round in 29.24 s, about 10.4× faster. Object detection is accelerated by about 32×, and segmentation by about 3×, decreasing to 0.67 s. However, MS–RGB fusion remains the main bottleneck, requiring 24.70 s, i.e., about 84% of the Jetson Orin runtime, showing that GPU acceleration reduces the overall runtime but does not eliminate the cost of image alignment and fusion. Under GPU execution, energy increases to 1.55 Wh, with 53.7 °C peak GPU temperature and 1.47 GB RAM usage.

We also benchmark the soil segmentation step on the training RTX 4090 GPU workstation and the Jetson Orin Nano to estimate deployment latency before UAV execution. For patch ratios of {0.04, 0.06, 0.08, 0.10} after warm-up, the workstation reaches 34/68/102/120 image-FPS, while the Orin Nano reaches 12/27/39/42 image-FPS, corresponding to average slowdown factors of about 2.8. Thus, for this workload, Orin latency can be roughly estimated as 2.8× the workstation latency, representing an interesting result for the practical deployment of this system.

Beyond runtime and resource usage, onboard edge processing also reduces the amount of imagery that must be transmitted over bandwidth-limited links. Starting from 40 MB of raw UAV imagery, onboard edge processing greatly reduces communication load: object detection cuts data transfer by 10× (to 4 MB), MS index mapping by 8× (to 5 MB), and patch classification by 5× (to 8 MB). By transmitting only processed outputs instead of raw multispectral/RGB data, the system enables efficient long-range UAV communications.

Finally, Figure 9 shows the Digital Twin output generated from the UAV mission data. It compares real UAV data with the reconstructed synthetic scene and depth information, illustrating the role of the DT in post-flight replay, visual validation, and synthetic data generation. In this prototype, the DT serves as an operational replication and analysis layer, rather than a closed-loop flight-control module.

V. CONCLUDING REMARKS

The integration of terrestrial and non-terrestrial networks can benefit from adopting UAVs as mobile communication and processing platforms for deployment in remote areas where these services are needed. We consider smart agriculture and environmental monitoring scenarios in which UAVs can be

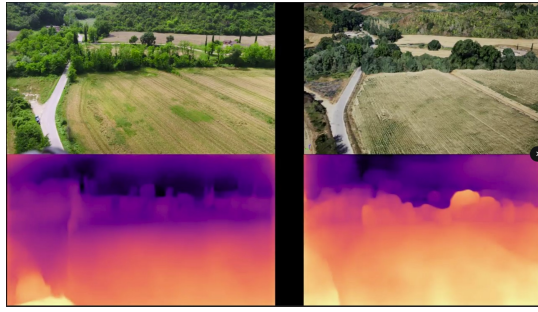


Fig. 9. Comparison between real data (on the left) and data generated by the UAV Digital Twin (on the right) with depth analysis.

remotely connected, also adopting a satellite system. This work demonstrates a novel end-to-end UAV-aided monitoring ecosystem that seamlessly integrates advanced sensing, edge AI processing, and 5G-enabled digital twin synchronization. The system has successfully validated the real-time execution of YOLOv11 detection, NDVI/NDRE mapping, MS-RGB fusion, and semantic segmentation on board under realistic operational constraints on processing capabilities and energy consumption. A key achievement of this system is the drastic optimization of communication resources. Instead of transmitting the full raw RGB/MS imagery, onboard edge processing converts sensor data into compact task-level outputs, such as detection metadata, vegetation-index maps, and semantic heatmaps. In this sense, the system reduces the transmitted data volume, achieving a 10-fold data reduction for object detection, an 8-fold reduction for multispectral mapping (NDVI/NDRE), and a 5-fold reduction for patch-based semantic classification.

Future work will focus on extending the proposed system to manage mission-critical scenarios, as addressed in the 5G-HUB project [3], and on considering scalable multi-UAV deployments, improving the efficiency of onboard AI processing, and supporting more complex monitoring and adaptive intelligence approaches.

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REFERENCES

- [1] R. P. Sishodia *et al.*, "Applications of Remote Sensing in Precision Agriculture: A Review," *Remote Sensing*, vol. 12, no. 19, p. 3136, 2020.
- [2] Y. Yazid, I. Ez-Zazi, A. Guerrero-González, A. El Oualkadi, and M. Arioua, "UAV-Enabled Mobile Edge-Computing for IoT Based on AI: A Comprehensive Review," *Drones*, vol. 5, no. 4, 2021.
- [3] 5G-HUB Consortium, "5G-HUB: Fifth Generation Services HUB for European Union Governmental Satellite Communications," <https://www.5g-hub.eu/>, 2026, accessed: 2026-05-06.
- [4] M. Masduzzaman, R. Nugraha, and S. Y. Shin, "IoT-based CO₂ Gas-level Monitoring and Automated Decision-making System in Smart Factory using UAV-assisted MEC," in *2022 International Conference on Decision Aid Sciences and Applications (DASA)*, 2022, pp. 1023–1027.
- [5] Tilon, Sofia and Nex, Francesco and Vosselman, George and others, "Towards Improved Unmanned Aerial Vehicle Edge Intelligence: A Road Infrastructure Monitoring Case Study," *Remote Sensing*, 2022.
- [6] Hernández, Daniel and Cecilia, José M. and Cano, Juan-Carlos and Calafate, Carlos T., "Flood Detection Using Real-Time Image Segmentation from Unmanned Aerial Vehicles on Edge-Computing Platform," *Remote Sensing*, vol. 14, 2022.
- [7] A. Koubaa, A. Ammar, M. Abdelkader, Y. Alhabashi, and L. Ghouti, "AERO: AI-enabled remote sensing observation with onboard edge computing in UAVs," *Remote Sensing*, vol. 15, no. 7, p. 1873, 2023.
- [8] "UAV-based estimates of corn LAI using hyperspectral and EnMAP spectral resolutions," *Computers and Electronics in Agriculture*, vol. 244, 2024.
- [9] H. M. Manoj, D. L. Shanthi, B. N. Lakshmi, K. J. Archana, E. Venkata Naga Jyothi, and K. Archana, "AI-driven drone technology and computer vision for early detection of crop disease in large agricultural areas," *Scientific Reports*, vol. 16, p. 2479, 2025.
- [10] A. Navarro, C. de Quinto, and J. A. Hernández, "Beyond visual line of sight: UAVs with edge AI, connected LLMs, and VR for autonomous aerial intelligence," 2025.
- [11] S. Bultmann, J. Quenzel, and S. Behnke, "Real-time multi-modal semantic fusion on unmanned aerial vehicles with label propagation for cross-domain adaptation," *Robotics and Autonomous Systems*, vol. 159, p. 104286, 2023.
- [12] C.-H. Huang, W.-T. Chen, Y.-C. Chang, and K.-T. Wu, "An edge and trustworthy ai uav system with self-adaptivity and hyperspectral imaging for air quality monitoring," *IEEE Internet of Things Journal*, vol. 11, no. 20, pp. 32 572–32 584, 2024.
- [13] Fondazione RESTART. (2026) S3-NETWIN: Network intelligence. Accessed: Mar. 13, 2026. [Online]. Available: <https://fondazione-restart.it/it/progetti/s3-netwin/>
- [14] A. Andreadis, G. Giambene, and R. Zambon, "Design and analysis of a LoRa-based system with scheduled transmissions from IoT nodes to UAV in rural areas," *Ad Hoc Networks*, vol. 175, p. 103868, 2025.
- [15] M. Figaro, F. Rossato, A. Bonora, M. Giordani, G. Schembra, and M. Zorzi, "Experimental evaluation of a UAV-mounted LEO satellite backhaul for emergency connectivity," in *Proc. International Conference on Computing, Networking and Communications (ICNC)*, 2026, accepted for presentation.
- [16] P. Zhu, L. Wen, D. Du, X. Bian, H. Fan, Q. Hu, and H. Ling, "Detection and tracking meet drones challenge," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 11, pp. 7380–7399, 2021.
- [17] K. He, X. Zhang, S. Ren, and S. Jian, "Deep residual learning for image recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.
- [18] J. W. Rouse, R. H. Haas, J. A. Schell, and D. W. Deering, "Monitoring vegetation systems in the great plains with ERTS," in *Proceedings of the 3rd Earth Resources Technology Satellite Symposium*, Washington, DC, USA, 1974, pp. 309–317.
- [19] A. A. Gitelson and M. N. Merzlyak, "Signature analysis of leaf reflectance spectra: Algorithm development for remote sensing of chlorophyll," vol. 148, no. 3-4, pp. 494–500, 1996.
- [20] M. Carpentiero, G. Vivone, R. Restaino, P. Addesso, and J. Chanussot, "An optimization procedure for robust regression-based pansharpening," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–16, 2022.
- [21] P. Addesso, G. Vivone, R. Restaino, M. Carpentiero, and J. Chanussot, "Regression-based injection for hyperspectral pansharpening within a generalized laplacian pyramid framework," in *2024 14th Workshop on Hyperspectral Imaging and Signal Processing: Evolution in Remote Sensing (WHISPERS)*, 2024, pp. 1–5.
- [22] O. Barrero and S. A. Perdomo, "RGB and multispectral UAV image fusion for Gramineae weed detection in rice fields," *Precision Agriculture*, vol. 19, no. 5, pp. 809–822, 2018.
- [23] C. R. Harris *et al.*, "Array programming with NumPy," *Nature*, vol. 585, pp. 357–362, 2020.
- [24] A. Paszke *et al.*, "PyTorch: An imperative style, high-performance deep learning library," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019, pp. 8024–8035.
- [25] G. Bradski, "The OpenCV library," *Dr. Dobbs's Journal of Software Tools*, 2000.